



CEO PROJECT 2026 GROUP 22

PaySure

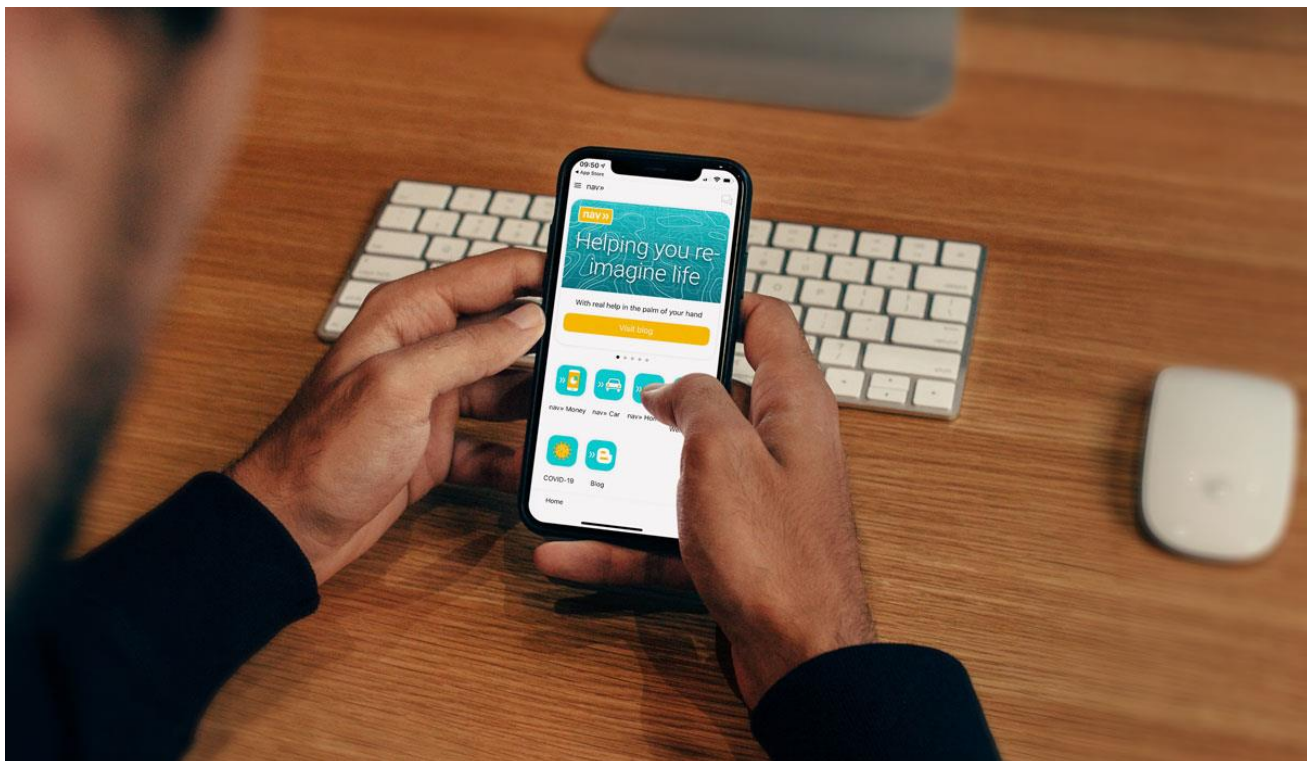


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1. INTRODUCTION

Debit order failures are commonly attributed to affordability constraints, yet a significant portion stem from timing frictions within the payment's ecosystem, cut-off times, sequencing delays, and balance availability gaps. Customers experience these failures as unpredictable and unfair, often incurring penalty fees and service disruptions despite intending to pay. The result is avoidable financial stress and erosion of trust.

FNB already has the foundational capabilities to address this problem—predictive cashflow modelling through nav»Money, real-time alerts via inContact, deep transactional data, and established credit facilities. However, these operate independently, leaving customers to interpret risks and react after failures occur.

PaySure integrates these capabilities into a unified predictive and decision-intelligence layer that prevents debit order failures before they happen. It enhances visibility around upcoming obligations and, where genuine short-term liquidity gaps are detected, recommends the most cost-effective funding path using existing facilities. Credit is applied contextually and time-bound, ensuring responsible intervention rather than indiscriminate expansion.

By shifting from reactive notification to proactive protection, PaySure reduces failed debit orders, protects product revenue, generates event-based liquidity income, and strengthens daily engagement within the FNB ecosystem. It is not a new product, but a strategic orchestration of existing infrastructure into a proactive financial safety layer.

2. MEET THE DREAM TEAM



**ALKA
SEWRAJ**



**MORRIS
NKOMO**



**MULISA
MATODZI**



**NEIL
SMIT**



**TAAHIR
KOLIA**

3. HIDDEN FAILURE IN EVERYDAY BANKING

3.1 CUSTOMER NEEDS

The critical blind spot in how customers understand, and experience debit order failures is that insufficient funds are rarely the sole culprit. While this remains the assumed cause, the reality is far more complex: a significant portion of failures stem from structural timing frictions embedded within the payment's ecosystem itself. Settlement delays, cut-off times, the sequencing of multiple debits, and the persistent gap between ledger and available balances all create narrow windows in which a customer may appear financially liquid yet still see a transaction fail.

This creates a significant visibility gap for customers, who are left with little clarity on why payments sometimes succeed and sometimes fail, often leaving them surprised by the outcomes. These failures carry real and uneven consequences, disproportionately affecting essential, high-value obligations like insurance premiums and loan repayments, categories where actual success rates fall materially below customer expectations. From the customer's perspective, the experience feels arbitrary and frustrating: funds may have been deposited, income scheduled to arrive the same day, and the intent to pay present, yet the payment still fails. The result is penalty fees, service interruptions, administrative friction, and a steady erosion of trust in their financial provider.

Within the FNB ecosystem, customers already have access to powerful but disconnected tools. Predictive balance modelling and real-time transaction alerts are available, but they operate as standalone features rather than an integrated system. This leaves customers to manually interpret dashboards, anticipate sequencing risks, and independently evaluate credit options when shortfalls emerge. The current approach reacts at the moment of failure rather than providing context-aware guidance before problems occur. This creates two fundamentally different scenarios that are unfortunately conflated in the customer experience: payment orchestration risk, where funds exist but system timing causes failure, and short-duration liquidity risk, where funds are genuinely unavailable within a narrow processing window. Today, both states trigger the same blunt response of a low balance notification followed by generic credit visibility or post-failure explanations.

What customers need is not simply more alerts, but clarity that accounts for the difference between timing risk and genuine shortfall and guidance that arrives before the failure, not after

3.2 THE SOLUTION GAP

nav»Money already detects when individual customers and SMEs are likely to enter short-term shortfall positions, based on cashflow patterns, upcoming debit obligations, and expected inflows. The bank therefore has visibility into liquidity risk before it materialises.

However, when these shortfalls occur, customers are typically presented with generic credit options (overdrafts, temporary loans, or facility increases) without contextual guidance on which option best fits their specific situation. These products are structurally sound, but they are not situationally intelligent. They are designed as standing facilities or broad liquidity tools, rather than responses tailored to a defined, time-bound shortfall event.

As a result, customers must independently evaluate complex trade-offs under time pressure: whether to allow a debit to fail, draw down on an overdraft, apply for new credit, or wait for incoming funds. This often leads to over-borrowing, incurring avoidable penalty fees, or entering facilities disproportionate to the duration and size of the gap.

For individuals, this can mean paying more in interest or merchant penalties than necessary, despite having predictable income inflows. For SMEs, even small shortfalls can disrupt payroll timing, supplier relationships, or service continuity, not because credit is unavailable, but because the most appropriate option is unclear at the moment it is needed.

The gap, therefore, is not the absence of credit mechanisms, but the absence of a decision intelligence layer that interprets shortfall context, repayment confidence, and product suitability and transparently recommends the most financially appropriate course of action.

4. RESEARCH

This section presents the empirical findings derived from the survey data collected to justify this study. The dataset, *Debit Order: The Early Bird That Took My Money.xlsx*, contains 29 complete responses and provides insights into customer experiences, challenges, and perceptions relating to the debit order processing. The findings directly support the identified problem space and validate the need for **PaySure**, a predictive overdraft and debit-order intelligence system.

4.1 PREVALENCE OF DEBIT ORDER FAILURES

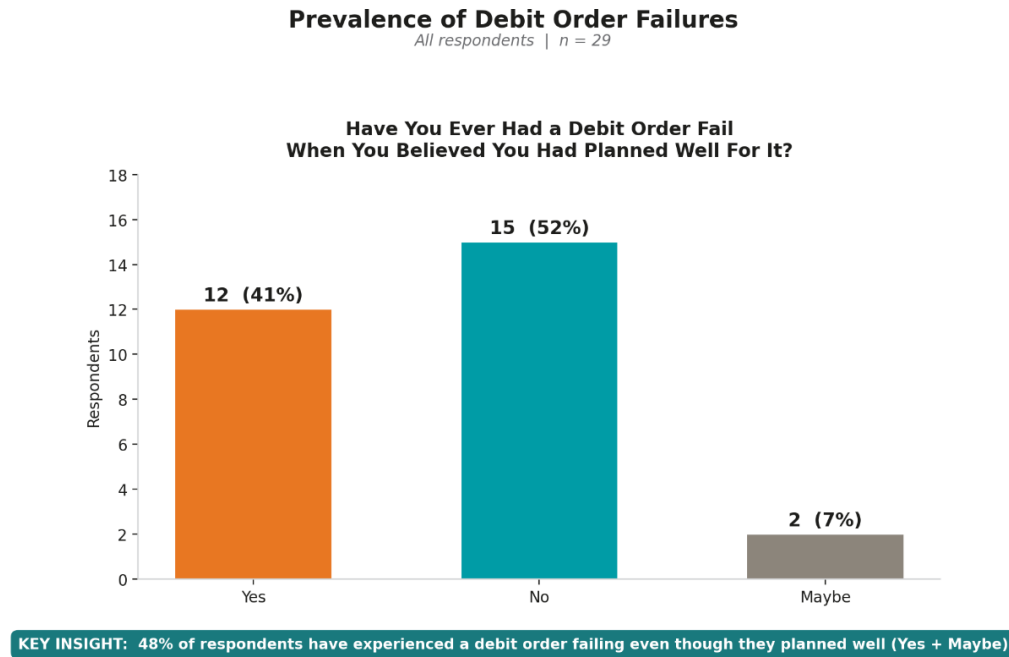


Figure 1: Prevalence of Debit Order Failures

From the findings, **48%** of respondents – nearly half – reported experiencing a failed debit order despite having planned adequately (**41%** responded “Yes”, and **7%** “Maybe”). In addition, many customers continue to encounter these failures with varying frequency: **41%** reported that failures occur “Rarely”, **17.2%** “Sometimes”, and **10.3%** “Often”.

These results show that debit order failures are not isolated incidents but a widespread and recurring issue across the customer base. This highlights the need for targeted interventions to reduce failure rates and improve the overall customer experience.

4.2 TIMING ISSUES AND VISIBILITY GAPS

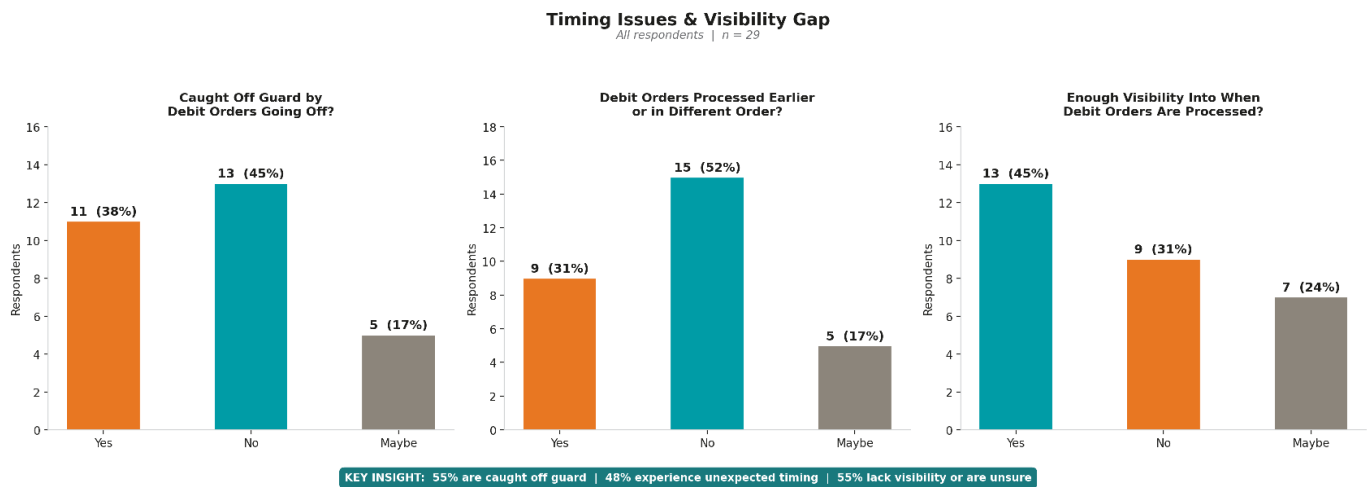


Figure 2: Insights into timing and visibility gaps

The survey revealed significant confusion and unpredictability around debit order timing:

- **38%** of respondents indicated they are caught off-guard by debit orders going off when they don't expect it.
- **31%** reported that debit orders sometimes go earlier or out of sequence.
- A combined **69%** of respondents either do not understand or are uncertain about why some debit orders go through and others do not.

Critically, visibility plays a substantial role in the customer experience. Among respondents who felt they did not have enough visibility, **66.7%** were caught off guard by debit orders. In contrast, only **15.4%** of those who felt they did have visibility experienced the same issue. This presents a **52** percent-point difference illustrating visibility as a key predictor of customer difficulties.

4.3 CATEGORIES OF DEBIT ORDERS MOST LIKELY TO FAIL

Multi-select analysis showed that the debit orders most prone to failure were:

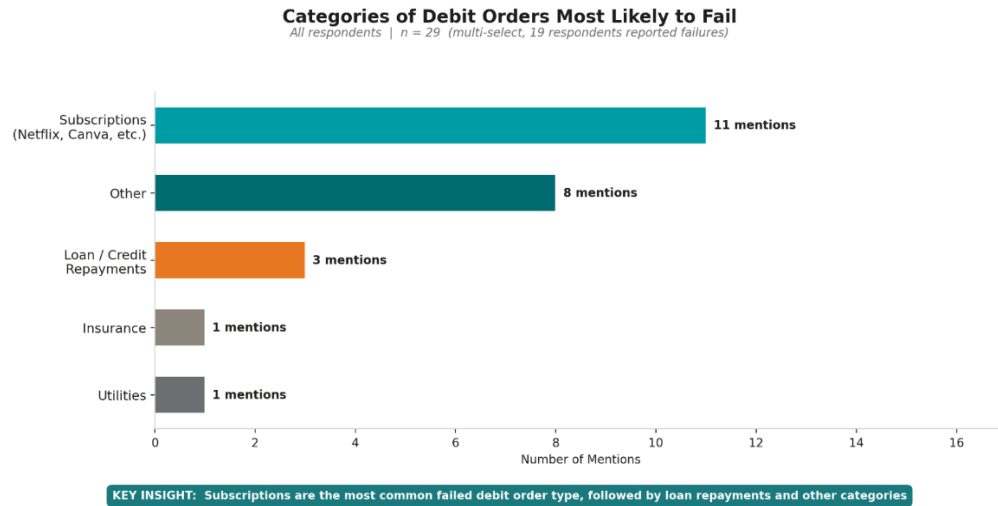


Figure 3: Categories of debit orders that fail

Although respondents in the survey primarily associated missed debit orders with subscription services, a separate dataset of transaction-level performance reveals that Insurance (**73.37% success rate**) and Loan Repayments (**79.20%**) are the most frequently missed debit orders in real-world processing. This discrepancy highlights a meaningful behavioural insight: consumers may underestimate failures in high-value categories while more vividly recalling unsuccessful subscription payments.

4.4 CONSEQUENCES OF DEBIT ORDER FAILURES

Participants reported experiencing multiple negative consequences following failed debit orders, as shown in Figure 4.

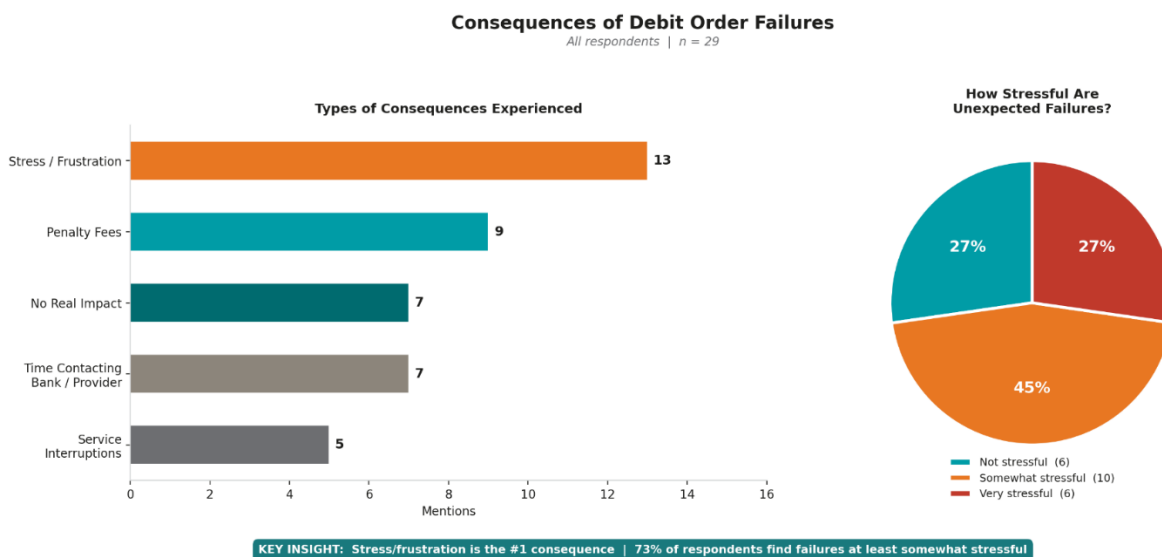


Figure 4: Common consequences of debit order failures

Furthermore, respondents who reported a debit order “bouncing” due to insufficient funds had a higher likelihood of incurring penalty fees (**35.7%**) compared to those who did not experience a bounce (**26.7%**).

4.5 SATISFACTION WITH BANK SUPPORT

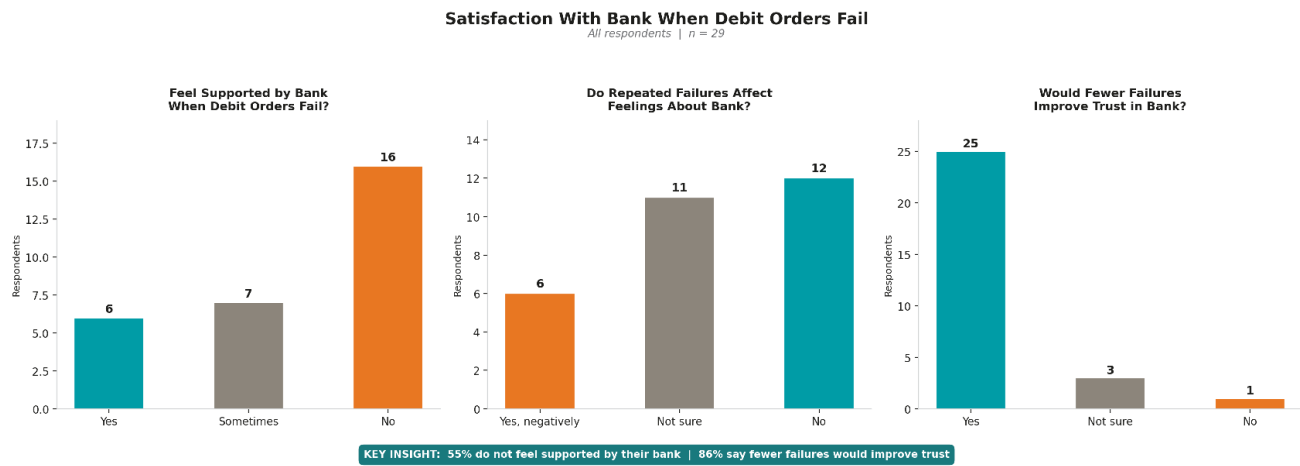


Figure 5: Customer satisfaction when orders fail

Most respondents, **55.2%**, reported that they do not feel supported by their banks when debit orders fail, while only **20.7%** felt supported. Additionally, **20.7%** indicated that repeated failures negatively affect their perception of their bank, with **37.9%** unsure. However, **86.2%** stated that fewer unexpected debit order failures would improve their trust in their bank, highlighting a meaningful opportunity for banks to restore customer confidence.

4.6 DEMAND FOR A PREDICTIVE DEBIT-ORDER SYSTEM AMONG FNB CUSTOMERS

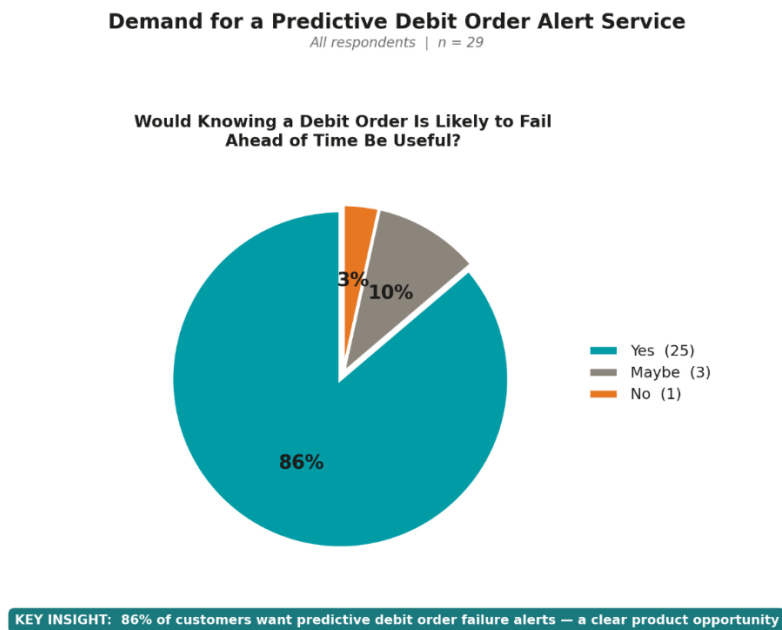


Figure 6: Insights into the need for more debit order visibility

To assess demand for a predictive debit-order-failure system specifically within FNB's customer base, survey responses regarding the usefulness of advance debit-order failure notifications were analysed alongside respondents' primary banking affiliations.

Across the full sample, **86.2%** of respondents indicated that advance notice of a likely debit-order failure would be useful, with only **3.4%** stating it would not be useful. When isolating responses by banking institution, a consistent pattern of strong demand emerges.

4.6.1 FNB CUSTOMER GROUP

FNB represented the largest customer segment in the survey, with 11 respondents identifying FNB as their primary bank (survey was random and not shared on any FNB associated groups). Within this group:

- 81.8% (9 out of 11) indicated that advance notifications would be useful.
- The remaining respondents indicated “Maybe”, with no FNB customers rejecting the usefulness of such a service.

4.6.2 COMPARATIVE INSIGHTS ACROSS BANKS

An analysis of all banks represented in the dataset shows that demand for predictive alerts is consistently high regardless of banking institution, but particularly pronounced among the major retail banks:

- Customers from **Absa, Capitec, Nedbank, and Standard Bank** all showed 100% usefulness agreement, although sample sizes were smaller.
- FNB customers showed an **81.8%** usefulness agreement, the highest among banks with a substantial number of respondents (n=11).

(Based on the expanded bank-response mapping from the dataset.)

This strongly suggests that FNB customers are just as motivated (if not more so) than other bank groups to adopt a predictive debit-order system.

4.7 INTERPRETATION FOR FNB

Two insights are particularly relevant for positioning PaySure within FNB.

First, there is a clear disconnect between high demand and low awareness. While the need for better debit order support is evident, most respondents (**58.6%**) indicated that their bank does not currently offer such a service, and a further 34.5% were unsure. Only 6.9% believed their bank already provided predictive debit order support. This points to a significant awareness gap and presents a strong opportunity for FNB to establish leadership in this space.

Second, FNB customers are experiencing the same pain points identified in the broader sample. Issues such as unpredictable timing, frequent debit order failures, and the resulting stress and frustration were commonly reported. These findings confirm that FNB clients face the same systemic challenges and stand to benefit directly from predictive interventions.

4.8 CONCLUSION

The combined analysis demonstrates that FNB customers show a strong and quantifiable appetite for a predictive debit-order-failure system, with more than four-fifths expressing support for such functionality. Demand is aligned with broader trends across all banks, but FNB's larger represented customer base provides clearer evidence of meaningful adoption potential. This reinforces the strategic viability of implementing PaySure within the FNB environment as a high-impact, customer-centric innovation capable of improving trust, reducing unexpected debit-order failures, and delivering proactive financial guidance.

5. THE SOLUTION: PAYSURE

Cashflow Assurance is a predictive financial layer that bridges the gap between customer intent and payment reality. It transforms reactive banking into proactive protection by continuously monitoring upcoming obligations, expected inflows, and real-time balance positions, distinguishing between timing risk, where funds exist but system friction causes failure, and genuine shortfall. Before a debit order fails, Cashflow Assurance simulates the impact of scheduled payments, surfaces upcoming commitments with clarity, and intervenes with contextual recommendations tailored to the specific gap, whether that means reshuffling payment timing, activating short-term liquidity, or accessing the most appropriate credit product.

What Cashflow Assurance is not is equally important. It is not simply another alert that waits for failure to explain what happened, nor is it a generic credit offer presented at the moment of stress. It is not a static dashboard requiring customers to manually connect dots between balances and dates, and it is not itself a credit product, it is a decision intelligence layer that interprets context and guides customers toward the most financially sound path. Critically, it is not one-size-fits-all: it accounts for the difference between a salaried individual with predictable inflows and an SME with irregular revenue cycles, tailoring guidance to the rhythm of each customer's financial life.

5.1 DEBIT ORDER ASSURANCE

Debit Order Assurance is the first of two integrated components that bring PaySure to life. Where the broader PaySure proposition distinguishes between timing risk and genuine shortfall, Debit Order Assurance addresses the former: ensuring that customers never fail a payment simply because they lacked visibility or time to act.

It transforms a passive banking experience into an actively protective one. Instead of waiting for a debit order to fail and then explaining why, the system intervenes beforehand, alerting customers to upcoming obligations, showing them the projected impact on their balance within the context of their broader cashflow, and giving them time to act. In doing so, it shifts customer behaviour from reactive crisis management to intentional financial planning, helping them avoid failed payments, penalty fees, and the downstream effects on credit health.

5.1.1 DATA ARCHITECTURE

The system would employ a hybrid data model combining verified mandate information with predictive analytics:

The system would leverage existing DebiCheck (DC), EFT, and Registered Mandate (RM) debit order data to deliver 100% accurate notifications for fixed bank-contracted payments. This ensures reliability for recurring obligations such as bond repayments, insurance premiums, and subscription services processed through formal mandate agreements.

For recurring payments without formal mandates (such as third-party subscription services), the system would utilise historical transaction data already powering nav»Money functionality. By analysing 6-12 months of payment patterns, the system would be able to accurately identify and flag recurring obligations with high confidence.

5.1.2 NOTIFICATION FRAMEWORK

Customers receive notifications two days before each scheduled debit order. This default timing balances advance warning with practical relevance, giving customers enough room to act without alerting them so early that the notification loses urgency. Timing is fully customisable, allowing preferences ranging from two to tens days' notice depending on the nature of the obligation, longer for rent or bond payments, shorter for entertainment subscriptions.

Notification delivery is designed for both reach and cost efficiency:

- **In-app push notifications**, integrated with inContact, serve as the primary and most cost-effective channel. They deliver alerts directly within the mobile banking app, where customers already engage with their finances.
- **SMS** is intentionally deprioritised for the standard consumer proposition due to cost considerations. However, it may be considered as an optional upgrade for Small Business Services (SBS) customers, where the higher reach and read rates of SMS could justify the expense, particularly if the solution generates measurable value, such as reduced failed payments or increased uptake of buffer facilities.
- **A future WhatsApp channel** is contemplated, conditional on the broader implementation of an FNB WhatsApp banking channel. If and when that foundation exists, customers could opt in to receive debit order notifications via WhatsApp, aligning with their preferred messaging environment while maintaining bank-grade security and authentication.

This channel strategy ensures that Debit Order Assurance remains accessible and financially sustainable, while retaining flexibility to serve segments where alternative channels deliver superior engagement.

5.1.3 ENHANCED NAV»MONEY DASHBOARD

UPCOMING OBLIGATIONS VIEW

It is proposed that a new section within nav»Money be added to provide customers with a consolidated list of all debit orders due (Fig. 7). While debit order mandates are currently visible in the profile section of the mobile banking app, this enhanced view delivers a contextual, action-oriented presentation specifically designed for cashflow management and budgeting purposes.

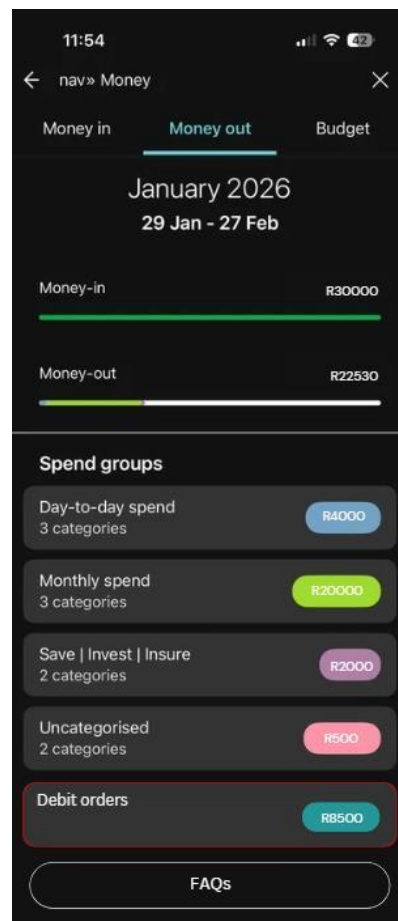


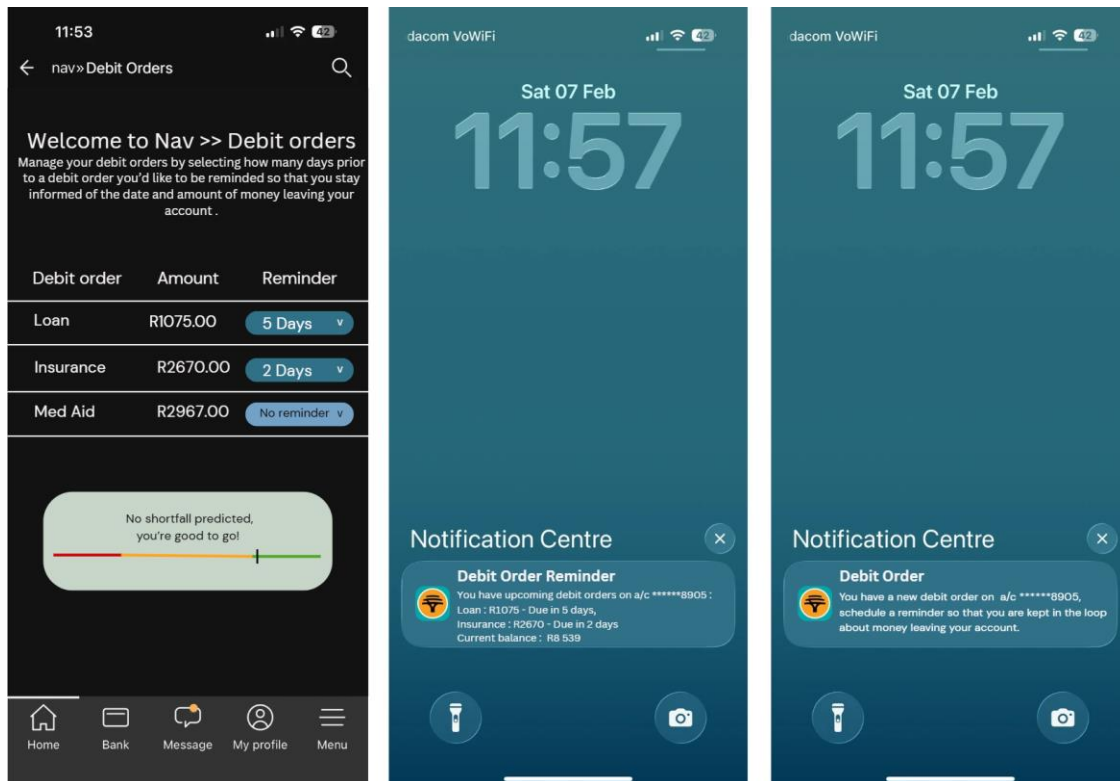
Figure 7: Debit order spend group integrated into nav>>Money page

DASHBOARD CAPABILITIES

From the nav >> Debit Orders page, clients can:

- Customise notification timing (2-, 5-, 7- or 10-day prior notice) for individual debit orders, the client may also opt for no reminder (Fig. 8.1)
- View projected balance impact showing whether customers would have a shortfall to cover the upcoming debits (Fig. 8.1 vs Fig. 9.1)
- The client will receive notifications to remind them of the debit order as per the reminder schedule that they had selected. The notification advises on the account that the debit order is scheduled to go off, reminders of upcoming debit order payments, as well as the current balance of the bank account which would give the client an understanding of whether they would have adequate funds for the debit orders to go through (Fig. 8.2).
- Should a new debit order be authorised, the client would receive a notification that a new debit order has been placed on the account and would be asked to schedule a debit order reminder (Fig. 8.3), the client may then select if and when they would like to be reminded about the new debit order (Fig. 9.1 to Fig. 9.3)

Figure 8: Mock up of the nav>>Money debit order page

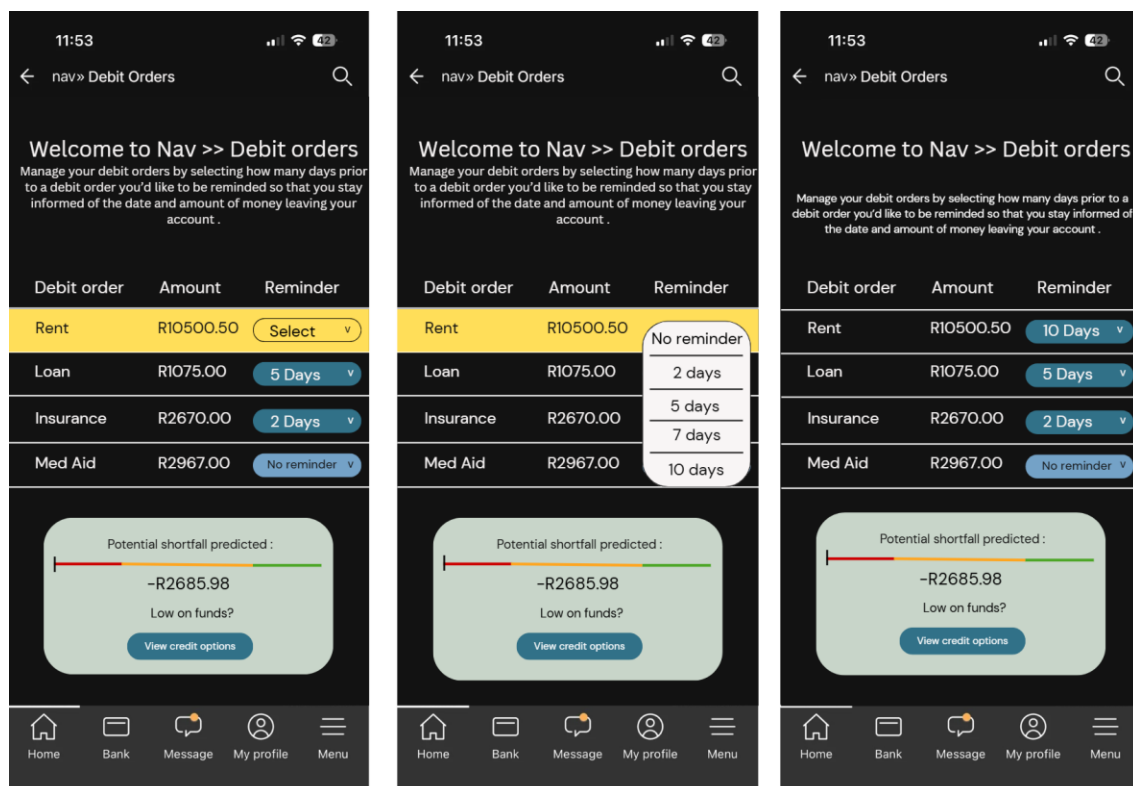


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8.3

Figure 9: Mock up of the inContact notifications for new and upcoming debit orders



9.1

9.2

9.3

5.1.4 COMPETITOR ANALYSIS

A competitor analysis was conducted to assess how major banks competing with FNB integrate their money management (MM) tools with transaction notification systems, with a particular emphasis on pre-emptive debit order alerts. The objective is to understand how effectively each institution enables customers to proactively manage their finances through a combination of predictive insights, budgeting capabilities, and real-time alerts. By examining the functionality, strengths, and shortcomings across these competitors, this analysis highlights areas where FNB maintains a competitive advantage, as well as opportunities for enhancement. These insights ultimately support FNB in strengthening customer financial awareness and driving deeper engagement with existing products

Table 1: Comparison of competitor's money management tools and notification frameworks

BANK	MONEY MANAGEMENT (MM) TOOL	MM TOOL OVERVIEW	TRANSACTION NOTIFICATION SYSTEM	PREDICTIVE DEBIT ORDER ALERTS	DISTINGUISHING MM OR NOTIFICATION FEATURE	GAPS
FNB	nav>>Money	Tracks & categorises spend; supports auto & manual budgets; predicts safe to spend and month end balances	InContact – SMS & in-app alerts for real-time transactions, with account balances included	No - real-time only	Predictive balance forecasting that can factor in future payments like debit orders and recurring commitments	Missing an awareness / proactive element; users must manually check dashboards on a continuous basis to stay ahead
DISCOVERY BANK	Financial Analyser	Categorises spend; supports manual & automated budgets; limited predictive financial insights	Email & in-app alerts for real-time transactions, with account balances included	Yes – Mandated debit orders	Offers multiple debit order alerts that notify users of upcoming debit amounts and account balances before debits-order alerts that notify users of upcoming debit amounts and	Missing deeper predictive budgeting and cash-flow projections
CAPITEC	None	Not Applicable	SMS alerts for real-time transactions.	No - real-time only	None identified	Overall, lacks money management insights.
ABSA	ABSA Savings Coach	Personalised savings goals; automated saving and progress tracking	NotifyMe – SMS, email & in-app alerts for real-time transactions, with account balances included app alerts time	No - real-time only	Users can opt-in for daily or weekly balance updates	Lacks insight into spend behaviour.
STANDARD BANK	Money Movements, Future Payments & Budget Manager	Tracks spend & inflows; supports manual budgeting; shows expected payments over 14- and 30-day periods	MyUpdates – SMS & in-app alerts for real-time transactions, with account balances included	No - real-time only	Dashboard surfaces expected payments at multiple levels (summary totals to individual upcoming payments)	Predictive dashboard requires manual addition and lacks visibility. Users are not pushed with proactive notifications
NEDBANK	MoneyTracker	Tracks spend & inflows; supports manual budgets; no predictive forecasting	SMS & in-app alerts for real-time transactions	Unclear	None identified	Missing predictive MM elements and notification features

From the competitive landscape, it is evident that FNB leads in predictive budgeting through features such as 'My Available Funds', which incorporates expected payments into a customer's projected balance. However, the transparency behind which payments are included in this calculation is limited. In contrast, Standard Bank's Future Payments feature offers strong visibility into upcoming transactions, presenting them clearly even though it lacks FNB's predictive balance capability. Additionally, introducing features like Discovery Bank's proactive debit order notifications could further enhance the nav»Money experience and significantly strengthen FNB's position in empowering customers with timely, actionable financial insights.

5.2 CONTEXTUAL CREDIT ENGINE (CCE)

The Contextual Credit Engine (CCE) is a decision-intelligence layer designed to optimise credit guidance at the precise moment a shortfall is predicted. It forms the second core component of the PaySure proposition, operating downstream of Debit Assurance and upstream of customer action. Where Debit Assurance addresses payment orchestration risk, ensuring that existing funds are marshalled effectively across timing frictions, the CCE addresses short-duration liquidity risk: genuine funding gaps that require financing to prevent payment failure.

Unlike traditional credit decision systems, which assess eligibility in isolation, the CCE evaluates credit suitability within a defined event context: a specific upcoming obligation, a quantified shortfall amount, and a measurable recovery window. Its purpose is not to originate new credit products or expand balance sheet exposure indiscriminately. Rather, it leverages existing FNB credit facilities and matches them intelligently to short-duration liquidity events. In doing so, it transforms credit from a static facility into a situationally optimised tool. The engine sits downstream of nav»Money's shortfall detection capability and upstream of customer interface guidance, bridging predictive visibility with responsible credit pathing.

5.2.1 DATA ARCHITECTURE

The CCE integrates four core capability layers already present within FNB:

- **Predictive Cashflow Modelling** – detects upcoming shortfalls based on expected inflows and scheduled debits
- **Credit Infrastructure** – overdrafts, temporary loans, credit cards, facility increases
- **Risk and Affordability Frameworks** – internal behavioural scoring and affordability models
- **Customer Interface Layer** – app-based guidance and simulations

Rather than replacing these systems, the CCE orchestrates them. Where traditional credit decisioning answers the question of whether a customer is eligible for a given product, the CCE answers a fundamentally different question:

“Given this specific shortfall event, which available facility results in the lowest cost, lowest risk, and highest repayment certainty?”

This shift from product-centric eligibility to context-centric optimisation is the defining innovation.

5.2.2 MODELLING FRAMEWORK

The CCE operates on a three-stage modelling structure.

Stage 1: Event Classification Model

In the first stage, the engine determines the nature of the predicted shortfall before any credit comparison occurs. It classifies the shortfall into one of two primary categories, as shown in Table 2.

This classification is critical because it determines whether credit guidance is appropriate at all. If orchestration risk can be resolved through Debit Assurance mechanisms alone, credit recommendation is suppressed. The classification model draws on expected inflow timestamp proximity, historical income consistency, past shortfall duration patterns, mandate retry behaviour, and balance volatility metrics. Only when liquidity risk is confirmed does the engine progress to product comparison.

Table 2: Shortfall classification scenarios

CLASSIFICATION	DESCRIPTION
Payment Orchestration Risk	Funds exist within a near-term window, but sequencing or settlement mechanics create a temporary deficit.
Short-Duration Liquidity Risk	Funds are genuinely unavailable within the debit processing window, requiring financing to prevent failure.

Stage 2: Repayment Certainty Assessment

In the second stage, the CCE calculates a Repayment Certainty Index (RCI) for the defined exposure window once liquidity need is validated. Unlike traditional credit scoring, which evaluates long-term default probability, the RCI focuses on short-horizon recovery likelihood. It synthesises the following inputs, shown in Table 3.

Table 3: Input variables that are synthesised

INPUT VARIABLE	DESCRIPTION
Income Stability Score	Coefficient of variation of net monthly inflows over a rolling window
Historical Short-Term Credit Repayment Performance	Proportion of short-term facilities repaid within agreed terms
Debit Order Success Ratio	Proportion of scheduled debits successfully honoured over a rolling period
Overdraft Utilisation Discipline	Average utilisation as percentage of limit; frequency of excesses
Savings Buffer Ratio	Average available balance relative to monthly outflow volatility

The RCI outputs a probability-weighted recovery confidence score over the specific shortfall duration, whether three days, seven days, or fourteen days. Tier segmentation guides decisioning is given in Table 4. This ensures that the engine does not recommend facilities where repayment timing confidence is weak.

Table 4: Tier segmentation guides decisioning

RCI TIER	SCORE RANGE	CREDIT DECISIONING
TIER 1	≥ 0.85	Full credit options available
TIER 2	0.70 – 0.85	Core credit options available
TIER 3	0.55 – 0.70	Restricted options, higher scrutiny
TIER 4	< 0.55	Credit suppressed; alternative guidance provided

The CCE includes embedded guardrails to ensure responsible operation, which are listed in Table 5. Together, these safeguards ensure the engine remains corrective rather than enabling, supporting customers through temporary gaps while protecting them from structural over-reliance.

Table 5: CCE guardrails

SAFEGUARD	FUNCTION
Repeated Shortfall Detection	Triggers structural review rather than continued liquidity guidance
Low RCI Tiers	Suppress high-risk facility recommendations
Duration Thresholds	Prevent long-term compounding risk
High-Frequency Usage Flags	Signal possible affordability deterioration

Stage 3: Contextual Product Optimisation Model

In the third stage, the engine conducts a comparative analysis of all eligible facilities available to the customer. For each option, it simulates the full cost and risk profile, factoring in draw amount, interest, fees, compounding risk, early repayment costs, and a behavioural premium tied to the customer's RCI tier.

From this, the engine calculates a Total Cost of Liquidity (TCL) and then adjusts for repayment uncertainty to derive a Risk-Adjusted Liquidity Cost (RALC). Each facility is then scored based on four key criteria: lowest projected cost, strongest alignment with repayment certainty, minimal behavioural risk, and lowest long-term dependency. The facility with the optimal combined score becomes the recommended path.

To prevent structural overuse, a behavioural dependency penalty is applied when shortfall frequency exceeds a calibrated threshold, initially set at three events over a ninety-day period. The final recommendation is the facility that minimises the sum of its risk-adjusted cost and this penalty, while respecting facility limits, RCI thresholds, and usage guardrails.

The engine also supports transparency by exposing a breakdown of this analysis to the customer, offering clear comparisons of costs over time and estimates of repayment confidence based on observed income patterns.

5.2.3 CREDIT FACILITIES UTILISED

Table 6 summarises the optimal use cases for each facility type.

The CCE transforms FNB's existing predictive capability into contextual financial intelligence. Competitors may offer alerts, budgeting tools, or credit products, but few integrate predictive shortfall detection with comparative credit optimisation in real time. The CCE therefore positions FNB not merely as a credit provider but as a liquidity decision partner. This moves the bank from reactive product distribution to proactive financial guidance; a distinction competitors cannot easily replicate without comparable predictive infrastructure.

Within the PaySure proposition, the CCE completes the arc: Debit Assurance prevents failures where funds exist but timing falters, while the CCE intelligently bridges genuine gaps where funds are absent, but intent remains. Together, they form a unified layer that protects customers from the hidden failures of everyday banking.

Table 6: Optimal use cases for different credit facilities

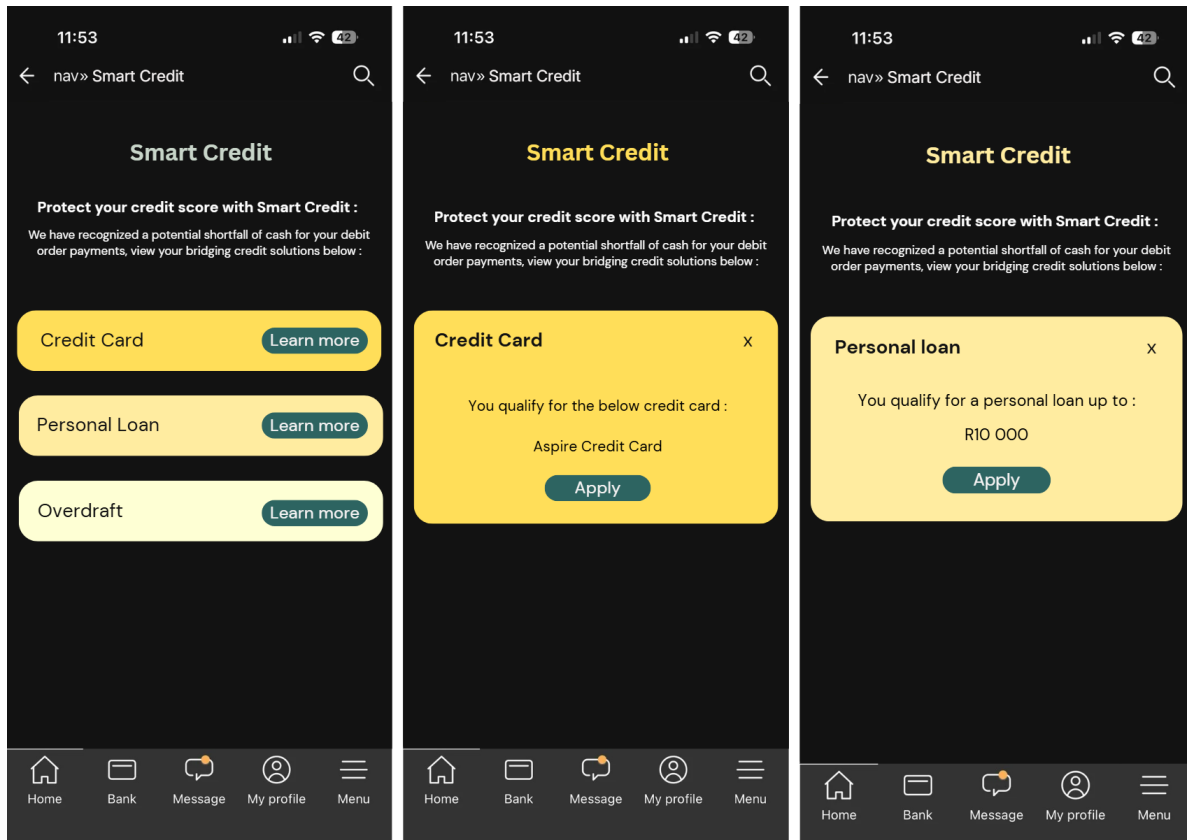
FACILITY	OPTIMAL USE CASE	RATIONALE
Overdraft	Very short duration gaps (1–5 days); customers with strong RCI scores; low draw amounts relative to income	Immediate liquidity; no new application friction; typically, lowest cost for short horizons
Credit Card Utilisation	Customers with unused credit limits; merchant-specific debit obligations; short-cycle repayment confidence	Interest-free period where applicable; no additional facility activation; repayment flexibility
Temporary Loan	Larger shortfalls; lower RCI tiers; shortfalls exceeding typical overdraft duration	Fixed repayment terms; predictable cost structure; reduced compounding uncertainty
Salary Advance	Short-term income mismatches; customers needing early access to a portion of confirmed upcoming salary	Provides structured early-access liquidity; reduces reliance on revolving credit; aligns repayment with next salary date

5.2.4 DESIGN MOCK SCREENS

Following from nav >> Debit Orders, once the client selects the button “View credit options” (Fig. 9.3), the client will be directed to the nav >> Smart Credit screen where their credit options will be visible (Fig. 10.1).

The client may click “Learn more” on each credit option to learn about the type of credit they qualify for and then apply for the credit that would satisfy their situation the most (Fig. 10.2, Fig. 10.3).

Figure 10: Mock up of the front end of the CCE



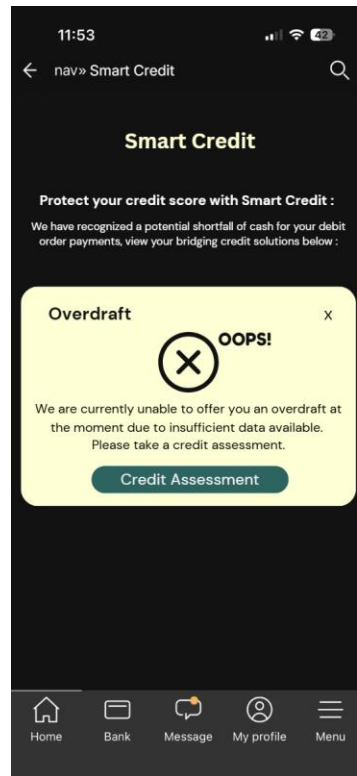
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10.2

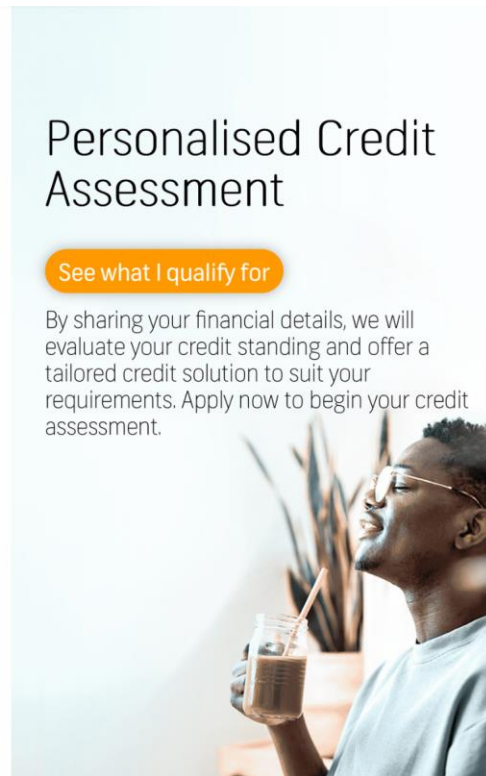
10.3

Should there be insufficient data on the client to provide them with credit, the client would be able to take a credit assessment, which is already offered on the FNB App, however this would allow the client to directly access this facility from nav >> Smart Credit (Fig. 11.1 & Fig. 11.2)

Figure 11: Mock up of when there is a lack of information on a customer



11.1



11.2

6. ABUSE & RISK PRESSURE TESTING

Any system that influences financial behaviour must be tested not only for its intended effects, but for how it behaves under pressure. The Contextual Credit Engine operates at the intersection of human behaviour, market dynamics, and macroeconomic forces. This section examines the key risks that could challenge its performance and the safeguards that keep it resilient.

6.1 BEHAVIOURAL GAMING

Customers learn how the system works and adapt. They might temporarily inflate balances before debits, repeatedly rely on short-term funding, or time transactions to access lower-cost facilities. These behaviours are economically rational, not malicious, but they distort the engine's signals.

The risk scenario.

A customer shifts funds in before debit processing and out again afterward, repeatedly triggering the lowest-cost recommendation. The system rewards short-term manipulation rather than genuine financial health.

The safeguard.

The CCE evaluates pattern stability, not isolated events. Abrupt balance changes and repeated micro-shortfalls reduce repayment confidence scores dynamically. If a customer triggers contextual funding too frequently, the engine shifts from short-term recommendations to structural intervention, such as affordability review. The goal is not to deny liquidity, but to prevent dependency loops.

6.2 DEPENDENCY AND CONDUCT RISK

If the engine consistently rescues customers from shortfalls, they may begin to treat contextual liquidity as part of their expected balance. This invisible credit creep reduces financial discipline and increases long-term risk exposure.

The risk scenario.

A customer relies on contextual funding for small gaps every month, never noticing the cumulative cost or recognising that their cashflow pattern signals a deeper problem.

The safeguard.

The CCE requires explicit cost comparison before any credit activation. It shows cumulative usage over rolling periods. It introduces graduated friction after repeated reliance. And it maintains a clear distinction between timing mismatches, where funds exist, and genuine affordability gaps, where they do not. This prevents the engine from masking structural stress with repeated micro-interventions.

6.3 CORRELATED LIQUIDITY STRESS

Shortfalls cluster around month-end, seasonal spikes, and economic shocks. If thousands of customers trigger contextual funding simultaneously, the engine becomes a large-scale short-term funding channel, straining liquidity buffers.

The risk scenario.

A month-end confluence of salary delays and high debit volumes sees twenty percent of eligible customers trigger funding at once. Portfolio exposure spikes faster than anticipated.

The safeguard.

The CCE operates under dynamic portfolio constraints. During systemic stress, repayment thresholds rise, funding limits compress, and preference shifts toward facilities already capitalised, such as approved overdrafts. The engine expands in stable environments and tightens under pressure, aligning with prudential risk principles.

6.4 MODEL DRIFT

Payment systems evolve. Retry logic changes. Customer behaviour shifts. A model calibrated on last year's patterns quietly degrades.

The risk scenario.

A change in industry retry windows means debits now retry two days later. The engine, unaware, continues recommending one-day facilities, and customers face unexpected second-day shortfalls.

The safeguard.

The CCE continuously back tests against realised outcomes, monitors prediction error dispersion, and recalibrates regularly. Qualitative review complements quantitative monitoring: if customer support tickets spike around specific recommendations, model assumptions are interrogated. Automation without governance is fragility.

7. FINANCIAL & STRATEGIC IMPACT

PaySure should be evaluated not as a feature enhancement, but as an infrastructure layer that reshapes debit-order economics. Its impact emerges from incremental improvement applied to very large volumes. Even marginal uplifts in success rates create meaningful cost, revenue, and capital effects.

The economic case rests on three measurable levers: reducing avoidable failures, protecting underlying product revenue, and activating short-duration liquidity in a controlled, capital-efficient manner.

7.1 OPERATIONAL BASE

Between April 2024 and January 2026, the bank processed over 233 million debit orders, of which 187.6 million were successful. The aggregate success rate of 80.46% implies a structural failure rate of 19.54%.

Table 6: Volume of debit orders processed by FNB from April 2024 to January 2026, with accompanying success rates.

METRIC	VOLUME
Submitted	233,211,673
Successful	187,642,515
Success Rate	80.46%
Failure Rate	19.54%

This means nearly one in five debit orders fails. Not all of these failures represent structural affordability distress. A portion reflects timing friction, sequencing gaps, or short-duration liquidity mismatches - precisely the layer PaySure is designed to address.

7.1.1 CHANNEL CONCENTRATION MATTERS

Table 7: Distribution of debit orders based on mandate type

CHANNEL	SUBMITTED	SUCCESSFUL	RATE
DC	85,431,900	75,131,922	87.94%
EFT	145,888,796	111,094,059	76.15%
RM	1,890,977	1,416,534	74.91%

EFT currently dominates transaction volumes and consequently carries the largest absolute number of failures. A possible reason for this is that EFT mandates are typically scheduled to run later in the day, combined with the fact that unsuccessful debit orders are not tracked for subsequent collection attempts. Strategically, this makes EFT a high-impact area where even marginal improvements could generate disproportionate financial returns.

Note: Registered mandates have only been around since 2025/06.

7.2 COST AVOIDANCE

Table 8: Overview failure rates across mandate channels

SEGMENT (2025/01 - 2026/01)	CURRENT FAILURES	1% FEWER	3% FEWER
Total (All Channels)	12,293,780	122,938	368,813
DC	2,404,484	24,045	72,135
EFT	9,426,499	94,265	282,795
RM	462,797	4,628	13,884

EFT dominates volume and carries the largest absolute failure base. Strategically, this is where marginal improvement yields disproportionate financial return.

Cost attachment formula:

$$\begin{aligned}
 &\text{Cost_Avoided (R)} \\
 &= \text{Avoided_Failures} \times [\text{Avg Penalty Fee per fail (R)}] \\
 &+ \text{Avoided_Failures} \times [\text{Avg Call/Back-office Cost (R)}] \times [\% \text{ of fails escalating}] \\
 &+ \text{Avoided_Failures} \times [\text{Avg Reinstatement/Admin Cost (R)}] \times [\% \text{ of impacted services}]
 \end{aligned}$$

Tip: If Finance wants the fastest quantified win, price the EFT rows first; that's where most savings will materialise.

7.3 REVENUE PROTECTION

Every avoided debit failure also stabilises revenue attached to the underlying product. These flows include loan instalments, insurance premiums, card repayments, and subscription-linked services

Revenue protection can be expressed as:

$$\begin{aligned}
 &\text{Revenue_Protected (R)} \\
 &= \text{Avoided_Failures} \times [\text{Avg Debit Amount (R)}] \times [\text{Product Margin \%}]
 \end{aligned}$$

The importance of this effect lies not only in current-period income preservation, but in preventing payment slippage that migrates into collections pipelines. Stabilising first-attempt success reduces arrears formation probability and improves portfolio health.

PaySure therefore functions as a revenue preservation mechanism as much as a notification enhancement.

7.4 CONTEXTUAL LIQUIDITY ACTIVATION INCOME

Where genuine short-duration liquidity gaps are identified, the Contextual Credit Engine activates an existing facility, based on repayment certainty and lowest risk-adjusted cost.

This is not credit growth driven by product push. It is event-bound liquidity alignment.

Income generation can be modelled as:

Exposure monitoring template:

$$\begin{aligned} & \text{Liquidity Income} \\ &= (\text{Avoided Failures} \times \text{Acceptance Rate}) \\ & \quad \times \text{Average Shortfall Amount} \\ & \quad \times (\text{Average Duration} / 365) \\ & \quad \times \text{Effective Yield} \end{aligned}$$

Because exposure is typically outstanding for only a few days and capped to the verified shortfall amount, the revenue per event is modest but capital efficient.

The strategic significance lies in the structure of the exposure. Duration is short, repayment is anchored to predicted inflows, and activation is filtered through the Repayment Certainty Index. This produces incremental income without structurally increasing long-term indebtedness.

7.5 BALANCE SHEET EXPOSURE (TIME-BOUND, CAPPED)

Unlike traditional lending expansion, Paysure exposure is event-specific and time-bound. The amount advanced is limited to the verified shortfall rather than full facility drawdown. Duration typically spans 24 to 72 hours and is anchored to expected inflow timing.

Exposure monitoring template:

$$\begin{aligned} & \text{EAD_PaySure (R)} \\ &= \Sigma [\text{Top-up Amount}_i \times (\text{Avg Days Outstanding}_i / 365)] \end{aligned}$$

Because duration is short and utilisation is capped, the risk-weighted asset impact remains controlled. Capital consumption is therefore materially lower than conventional credit growth strategies.

8. COMPETITIVE AND STRATEGIC POSITIONING

PaySure's defensibility does not lie in the idea of debit-order alerts or short-term liquidity support. Those ideas are replicable. Its defensibility lies in the integration of deep behavioural data, predictive modelling, credit infrastructure, and customer interface within a single banking ecosystem.

FNB is uniquely positioned to deliver this because PaySure relies on behavioural signals that only a primary transactional bank can observe at scale and over time. These signals are not surface-level balances; they are temporal patterns embedded in daily cashflow movement.

The system draws on intra-month income timing, debit-order sequencing across rails, balance volatility patterns, inflow predictability, utilisation discipline across credit facilities, and historical payment success ratios. The predictive accuracy of PaySure is therefore not abstract; it is trained on FNB-specific behaviour inside FNB's own operating environment.

Competitors may replicate components but without equivalent depth of transactional continuity and credit-pathing integration, they cannot replicate the predictive precision that makes PaySure commercially viable and risk-contained.

8.1 STRUCTURAL FIT WITHIN FNB'S ECOSYSTEM

PaySure does not require the invention of new infrastructure. It connects capabilities FNB already operates but currently runs in parallel rather than in concert.

InContact delivers real-time alerts. nav»Money forecasts balances and categorises spend. The bank already holds rich longitudinal transactional histories. Credit rails (overdrafts, temporary loans, credit cards) are fully operational. Internal data science infrastructure already supports behavioural scoring and affordability modelling.

PaySure links these elements into a closed loop: forecast emerging risk, surface it proactively, evaluate contextual liquidity need, and guide the customer to the lowest-risk resolution path. The innovation is orchestration, not invention.

This is strategically important. Many competitors struggle to unify money-management tools with credit decisioning or to align notification systems with predictive modelling. PaySure works precisely because these components already exist within FNB's architecture.

8.2 PREDICTION VERSUS VISIBILITY

Across the competitive landscape, most banks provide visibility. Customers can view balances, track spending categories, and sometimes see upcoming payments. But visibility is reactive; it depends on the customer interpreting information correctly and at the right time.

PaySure shifts the model from visibility to prediction. Instead of requiring customers to monitor dashboards, it anticipates a potential failure and intervenes before the debit window closes. The distinction is subtle but powerful. One model informs; the other protects.

This predictive layer materially deepens customer engagement because it integrates into daily financial life. It becomes part of how customers experience their salary cycle, not simply a feature they check periodically.

8.3 ECOSYSTEM EMBEDDEDNESS AND SWITCHING FRICTION

PaySure strengthens retention not through artificial lock-in mechanisms, but through embedded usefulness. Its value depends on the customer's live income patterns, debit-order behaviour, balance volatility, and credit usage within FNB.

The more PaySure learns from a customer's financial rhythm, the more precise its forecasting becomes. That precision is not portable. If a customer migrates primary banking elsewhere, the predictive model loses behavioural depth.

This creates a form of strategic embeddedness rooted in data continuity. The benefit is not contractual stickiness; it is behavioural dependency on predictive protection. That is materially harder for competitors to displace.

8.4 COMPETITORS CAN COPY THE IDEA, BUT NOT THE EXECUTION

While the idea of proactive debit support appears straightforward, its execution requires coordinated integration across multiple systems:

- Predictive balance modelling calibrated to intraday timing
- Debit-order risk scoring across payment rail
- Real-time shortfall simulation
- Contextual credit optimisation
- Behavioural risk filtering and affordability guardrails
- Multi-channel notification orchestration

Each of these layers is individually complex. Combined, they form a multi-system capability that requires aligned risk governance, technology architecture, and regulatory oversight. Most institutions struggle to integrate even two of these layers effectively. PaySure requires all of them to function coherently.

Its defensibility therefore lies in systemic coordination, not conceptual novelty.

8.5 REGULATORY AND GOVERNANCE CONSIDERATIONS

Because PaySure operates at the intersection of prediction, credit guidance, and automated intervention, governance design is critical.

First, responsible lending principles must remain central. The Contextual Credit Engine does not expand eligibility beyond existing credit frameworks. It optimises within pre-approved or policy-compliant facilities and is filtered through affordability and behavioural scoring models already aligned with regulatory standards. This ensures the system does not introduce incremental reckless-lending exposure.

Second, explainability is essential. Where automated guidance is provided, the system must surface transparent reasoning for example, highlighting expected inflow timing, shortfall magnitude, cost comparisons across facilities, and repayment confidence. This aligns with fair-treatment principles and mitigates algorithmic opacity risk.

Third, usage monitoring and guardrails prevent structural dependency. Repeated shortfall detection triggers review mechanisms rather than continued liquidity activation. This shifts the system from being an enabler of chronic borrowing to a detector of emerging financial stress. From a regulatory standpoint, this positions PaySure as protective rather than exploitative.

Fourth, data governance and consent frameworks must be explicit. PaySure operates on first-party transactional data within the bank's ecosystem. No external data scraping or cross-institutional data sharing is required. Customers should retain the ability to opt out of proactive liquidity suggestions while still receiving notification alerts, preserving autonomy.

Finally, automated decisioning oversight must be embedded within risk governance structures. Model validation, bias testing, stress scenario simulation, and periodic recalibration should form part of the operational framework. Given that the system is event-specific and short-duration in exposure, model risk remains bounded but must nevertheless be actively managed.

When governed correctly, PaySure aligns strongly with prudential objectives. It reduces payment failure, stabilises collections, improves repayment reliability, and detects early-stage affordability deterioration. In this sense, it supports systemic credit quality rather than undermining it.

8.6 PAYSURE PROS & CONS

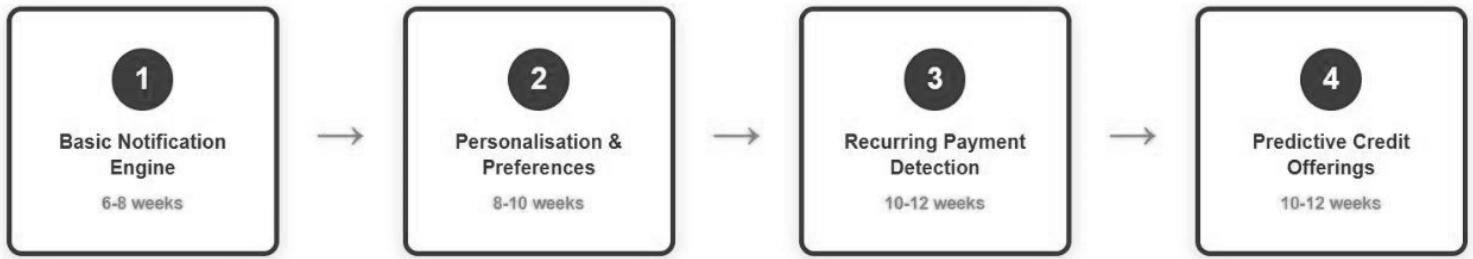
8.6.1 PAYSURE PROS

- Leverages existing infrastructure: Uses NAV, InContact, transactional data, and existing credit facilities.
Low build complexity relative to creating a new product.
- Reduces preventable revenue leakage: Fewer failed debits protect interchange, insurance, and partner income streams.
- Creates event-based liquidity income: Monetises short-duration shortfalls without expanding long-term exposure.
- Improves risk-adjusted returns: Applies liquidity only where predictive certainty of recovery is high.
- Strengthens customer trust and retention: Eliminates perceived “unfair” failures, increasing primary bank positioning.
- Enhances ecosystem stickiness: Increases daily app engagement through proactive financial guidance.
- Scalable intelligence layer: Creates feedback loops that continuously improve predictive accuracy.

8.6.2 PAYSURE CONS

- Model accuracy risk: Incorrect shortfall prediction could lead to unnecessary credit application or missed failures.
 - Behavioural risk: Customers may rely on PaySure support and reduce proactive cashflow management.
 - Regulatory and fair lending scrutiny: Automated, contextual credit application must meet transparency and affordability standards.
 - Operational complexity: Requires tight integration across payments, credit, risk, and digital teams.
 - Margin sensitivity: Short-duration liquidity may produce lower yield unless structured carefully.
 - Customer perception risk: If not clearly communicated, customers may misunderstand automatic intervention.
-

9. ROADMAP



This roadmap outlines a phased customer engagement optimisation journey that begins with a Debit Order Assurance notification system designed to enhance customers' financial management. From this foundation, the journey evolves toward generating incremental value to customers and simultaneously revenue for the bank through predictive credit offerings via the CCE. The four phases described provide a structured path from a minimum viable product to progressively advanced capabilities. By adopting a phased approach, the organisation can continuously improve the solution while gathering valuable customer feedback at each stage, ensuring that every enhancement aligns with real user needs and delivers measurable impact. Further details about each phase can be found below:

PHASE 1: FOUNDATION - BASIC NOTIFICATION ENGINE

DURATION: 6-8 WEEKS

Establish a minimum viable product (MVP) by using existing InContact notification infrastructure coupled with DC, RM, and EFT mandated debit order data, allowing customers to receive notifications via SMS and in-app push notifications without user controls.

KEY DELIVERABLES

- Registered debit order mandates (DC, RM & EFT) integration
- InContact SMS and in-app notification integration
- Backend scheduling & delivery APIs

RESOURCES REQUIRED

1 Product Manager, 1 Solution Architect, 2 Backend Developers, 1 Mobile Developer, 1 Quality Assurance (QA) Engineer, 1 DevOps Engineer

PHASE 2: PERSONALISATION & CUSTOMER CONTROL

DURATION: 8-10 WEEKS

Empower customers with granular control via dedicated nav>>Money debit order page, where they can view item level details around their mandated debit orders, with the option to configure notification preferences.

KEY DELIVERABLES

- Dedicated nav>>Money debit order page
- Individual listings of all mandated debit orders with notification configurations

RESOURCES REQUIRED

1 Product Manager, 1 UX/UI Designer, 2 Backend Developer, 2 Frontend Developer, 1 Mobile Developer, 1 QA Engineer, 1 Data Analyst

PHASE 3: RECURRING PAYMENT DETECTION

DURATION: 10-12 WEEKS

Comprehensive coverage beyond registered debit order mandates to include subscription services. This would require the deployment of machine learning (ML) models to identify non-mandated recurring subscription payments using 6-12 months transaction history and would make use of existing nav>>Money categorisation.

KEY DELIVERABLES

- ML pattern recognition for subscriptions
- Notifications for non-mandated debit orders and recurring payments
- Visibility of non-mandated debit orders on nav>>Money debit order page with configurability

RESOURCES REQUIRED

1 Product Manager, 1 Data Scientist, 1 ML Engineer, 2 Backend Developers, 1 Data Engineer, 1 QA Engineer, 1 Data Analyst

PHASE 4: PREDICTIVE CREDIT OFFERINGS & FINANCIAL INTERVENTION

DURATION: 10-12 WEEKS

This phase focuses on evolving the idea into a revenue-generating capability by preventing payment failures and enabling responsible credit growth. With the CCE, a status bar on the debit order page will indicate whether customers have sufficient funds to meet upcoming financial obligations. If a shortfall is detected, customers will be proactively presented with credit options. Customers will then be guided to the most suitable product.

KEY DELIVERABLES

- Build upon existing predictive cashflow modelling to contextually identify customers' potential cash shortfalls
- A credit product offering model
- Real-time credit pre-approval
- Balance status integration into nav>>Money debit order page

RESOURCES REQUIRED

1 Product Manager, 1 Credit Risk Analyst, 1 Data Scientist, 2 Backend Developers, 1 Frontend Developer, 1 Mobile Developer, 1 QA Engineer, 1 Compliance Officer

10. SUCCESS METRICS

The following measurable outcomes will determine program success and inform go/no-go decisions between phases:

	SUCCESS METRIC	TARGET
1	Customer Adoption Rate - Percentage of eligible customers actively receiving and engaging with notifications	≥50% by Month 6, ≥75% by Month 12
2	Failed Debit Order Reduction - Decrease in failed debit orders for customers receiving notifications	≥5% reduction in failed debit orders
3	ML Model Accuracy - Precision of recurring payment predictions.	≥95% accuracy for recurring payments
4	Credit Conversion Rate - Percentage of insufficient funds predictions that result in credit product uptake	≥10% conversion rate

APPENDIX A

A.1 INITIAL BRAINSTORMING & IDEA EXPLORATION

Our process began with a comprehensive audit of FNB's customer touchpoints, ranging from mobile and online banking to physical branches, ATMs, and backend systems. By casting a wide net across these diverse domains, we aimed to surface high-impact opportunities before narrowing our focus to the most critical pain points. Through this discovery phase, we determined that all identified challenges converged into three areas: Security, Operations, and Cash Flow Management.

DOMAIN 1: SECURITY

Problems and Ideas Explored:

- **Prepaid Accounts:** Virtual cards linked to primary accounts remain vulnerable in card not present transactions, and without isolated funding pockets or strict limits, a single compromise can expose the main account to fraud. Having the option to move money into prepaid pockets that are linked to virtual cards can add an additional security element for customers, especially when transacting online.
- **Session Management:** The FNB mobile banking app currently does not log users out when their phones lock while the app is active, creating a security risk in situations where a device is stolen during use. In such cases, a thief could immediately access the unlocked banking session. To mitigate this vulnerability, the app requires stricter session management controls that automatically log users out after a phone lock or after a short period of inactivity, ensuring that unauthorised access is prevented.

DOMAIN 2: CASH FLOW MANAGEMENT

Problems and Ideas Explored:

- **Proactive Notifications:** Customers often experience debit order failures not because they lack funds, but due to hidden timing frictions that make payment outcomes unpredictable and leave them confused and penalised. While tools like nav»Money's *My Available Balance* estimate what a customer's balance will be after expenses, they do not show which recurring payments are coming up or when to expect them. This reveals a gap in the proactiveness of current features. By notifying customers in advance about upcoming obligations and their expected balance at the time of each debit, the bank can help customers plan better, avoid surprises, and manage their money more proactively.
- **Contextual Credit Offers:** Additionally, when shortfalls are predicted, customers are presented with generic credit options and must independently decide whether to borrow, allow a debit to fail, or wait for inflows, often leading to overborrowing, unnecessary fees, and suboptimal financial decisions because they are not provided with event-specific context. It is proposed that a system be developed to interpret the customer's upcoming obligation, shortfall size, and repayment window to match them with the most appropriate existing FNB credit facility, ensuring precise, situationally intelligent liquidity support that prevents failures without increasing unnecessary exposure.
- **Predictive Overdraft:** Similarly, an additional idea explored was the provision of small, dynamically preapproved overdraft facilities that activate only when a shortfall is predicted, using the same contextual evaluation factors as the Contextual Credit Engine. This would give customers a flexible, event-specific buffer to prevent payment failures.

DOMAIN 3: OPERATIONS

Problems and Ideas Explored:

- **Customer Effort Index:** Customers often struggle to complete basic tasks, leading them to call, chat, or visit a branch only because the process didn't work smoothly. For example, repeated failures, confusing steps, or forced channel switching. These unnecessary help-requests increase frustration, repeat contacts, and operational workload, and the bank currently lacks data-driven visibility into which journeys create this friction. To solve this, a Customer Effort Index could be used to analyse interaction data, identify the journeys that force customers to seek help unnecessarily, and prioritise fixes that reduce friction, repeat contacts, and cost.
- **Customer Intent Self-Service:** Customers frequently visit branches or call centres for simple tasks that already exist on the banking app because they don't know where to find them, struggle with navigation, or experience login friction. This creates avoidable queues, higher operational costs, customer frustration, and inefficient use of staff time, especially when customers cannot express their needs in a way the system understands. The idea would be to introduce an intent-driven self-service layer in the app and in branches that lets customers describe what they want to do in plain English, automatically identifies if the task can be completed digitally, and guides them step-by-step on the app.
- **Rural ATM Access:** Parents in the Vhembe District can only make school payments through FNB cash deposit ATMs, but the area has very few deposit-enabled machines. This forces long travel, high costs, long queues, and frequent failed payment attempts, delaying learners' access to learning material and creating financial access barriers for rural families. This requires an increase in access to affordable payment options by expanding deposit-enabled ATM coverage, possibly partnering with local retailers for cash-in services, or offering simple USSD/mobile payment channels so parents can pay without travelling long distances.

A.2 IMPACT MATRIX & PRIORITISATION

To prioritise which domain to pursue, we developed a decision matrix evaluating each domain against seven criteria areas, which can be found in Table A1.

Table A1: Prioritisation matrix for different domains

Evaluation Criteria	Domain 1: Security	Domain 2: Cash Flow	Domain 3: Operations
Customer Breadth	Low Targeted to specific incidents (fraud, device theft).	High Daily, widespread debit-order & timing issues affecting majority of customers.	Low-Medium Operational friction common but variable across journeys.
Financial Impact on Customer	Low Protects funds but low ongoing value creation.	High Reduces penalty fees, arrears, and unexpected debit failures.	Low-Medium Saves time & repeat visits, but indirect financial effect.
Implementation Complexity	Low Feasible changes within app security & card architecture.	Medium Requires analytics, modelling, orchestration, credit logic. However, the existing app infrastructure is sufficient for an MVP.	High Cross-channel processes, data analysis, high effort infrastructure changes.
Time to Value	Moderate-Fast Fast fixes for session management; moderate for prepaid pockets.	Moderate Requires modelling, testing, staged rollout.	Slow Requires behavioural data, cross-functional delivery.
Measurability	Low-Medium Fraud incidents & session risk measurable but low volume for the latter.	High Clear metrics exist, such as debit success rate, unpaid fee reduction, shortfall conversion.	Low Many variables influence customer effort, making it a difficult attribution.
Revenue Impact on the Bank	Minimal direct benefits.	Short-term loss on collected penalty fees but long-term retention and lower credit impairments.	Indirect benefits.
Weighted Score	12/18	15/18	8/18

Key: Colour and
Associated Score

1

2

3

A.2.1 DISCUSSION

Using the impact matrix to refine our focus, we identified several topics that do not warrant further customer engagement at this stage. These items were deprioritised because they offer limited impact relative to the effort, overlap with existing capabilities, or fall outside the strategic scope. In short, they either address niche use cases, constitute hygiene work rather than strategic differentiation, or require long lead times and heavy investment with narrow benefit.

PREPAID DIGITAL CARD

This concept primarily targets customer anxiety around online security and spend control. However, most of the benefit can already be achieved through existing tools such as dedicated virtual cards, with per-card limits, and dynamic CVVs. As a result, the incremental value of a prepaid construct is low, and the audience is niche.

SESSION MANAGEMENT

This is an important security item that requires engineering remediation, but it is not a customer-engagement initiative. The issue is likely to have a relatively low incidence rate, and the solution is binary (secure or not secure), which limits strategic upside beyond closing a vulnerability. It should be fixed quickly with clear acceptance criteria, but it does not justify a deep dive into customer engagement optimisation.

RURAL CASH DEPOSITS

While the problem is real and important from a financial inclusion perspective, solving it demands substantial capital and operational effort, deployment of deposit-enabled ATMs, cash logistics, and retailer partnerships, on multi-year timelines. The benefit is geographically constrained compared to nationwide pain points that can be addressed sooner.

CUSTOMER EFFORT INDEX

CEI is a valuable diagnostic framework to identify friction points, but it is not a tangible customer deliverable. By itself, it does not create value unless paired with funded interventions and accountable. The right approach is to embed a CEI as instrumentation across programmes, using thresholds to trigger backlog items, rather than treat it as a standalone initiative.

CUSTOMER INTENT SELF-SERVICE.

Although intent-driven guidance has merit, it significantly overlaps with the upcoming NAVI capability. The NAVI chatbot is expected to translate natural language into actionable customer insights, which directly addresses the same need that a self-service layer would attempt to solve. Once NAVI is deployed, the idea ought to be revisited and assessed to understand whether it can provide the contextual, step-by-step direction customers require and whether products like NAVI can be extended and integrated into branch environments to support in-person self-service as well.

PREDICTIVE OVERDRAFT

FNB already offers several credit tools (overdrafts, loans, credit cards), and customers may struggle to understand when to use which. Adding a new “predictive overdraft” concept would increase choice overload and adoption friction without solving the core need.

SELECTED PRIORITY AREAS: DEBIT ORDER NOTIFICATIONS AND CONTEXTUAL CREDIT OFFERS

Cash flow management emerged as the strategic priority because customers experience failed debit orders frequently rather than hypothetically. Unlike future benefits, like enhanced security, or abstract improvements, cash flow shortfalls and failed debit orders represent an immediate, tangible problem causing financial stress in real time. Improving cash flow management and reducing failed debit orders would have a far wider reach than the use cases addressed by alternative domains.

A.3 INTERACTIVE LOOK INTO THE SITE

